

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.7	8.4	1	0.3	0.3	55.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.2	7.6	0.3	0.1	0.1	55.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.7	9	1.1	0.3	0.3	55.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.6	7.4	0.2	0.2	0.1	55.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.8	10.4	1.5	0.5	0.2	55.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.3	7.3	0.5	0.2	0.1	55.6
U2 - Efficacy additive alpha prior Beta(1, 1)	33	9.3	1.1	0.4	0.2	56
U3 - Efficacy additive alpha prior Beta(1, 1)	34.3	9.6	0.2	0.2	0	55.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	4.3	2.4	1	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.8	4.2	2.4	1	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.8	4.3	2.4	1	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.9	4.2	2.4	1	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.8	4.3	2.4	1	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.9	4.3	2.4	0.9	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	14.8	4.3	2.4	1	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	14.8	4.3	2.4	1	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.6	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.6	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.6	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.4	14.6	32.7	11.4	3.1	1.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	37.8	16.3	31.3	10.8	2	1.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.7	15.1	33	11.4	2.9	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	38.6	16.2	31.2	10	2.2	1.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.4	15.1	33	11.6	3	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	38.1	16.8	31.5	9.5	2.2	1.9
U2 - Efficacy additive alpha prior Beta(1, 1)	34.3	15.7	33.9	10.6	3.6	1.9
U3 - Efficacy additive alpha prior Beta(1, 1)	37.8	15.9	32	10.4	2.1	1.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	5.2	7.1	4.1	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	5.3	7.1	4.1	2.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	5.2	7.1	4.1	2.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	5.3	7	4.2	2.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	5.3	6.9	4.1	2.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	5.3	7	4	2.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.1	5.4	7	4.1	2.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	11	5.3	7.2	4.1	2.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.3	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.3	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.3	1.3	0.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.4	1.3	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.4	1.3	0.9	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.3	1.3	0.9	0.4	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.9	37.2	23.6	8.3	2	10
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.2	41	20.3	7	1.6	9.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.6	37.2	24.1	8.3	1.8	10
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	40.9	19.9	8.3	1.5	9.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.5	38.2	22.7	8.2	2.4	10
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	40.8	20.4	7.1	1.6	9.9
U2 - Efficacy additive alpha prior Beta(1, 1)	18.6	36.6	24.4	8.3	2	10.1
U3 - Efficacy additive alpha prior Beta(1, 1)	20.3	40.5	20.9	6.6	1.8	9.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	7.3	6.3	3.9	2.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	7.3	6.3	3.9	2.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.3	6.2	3.9	2.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	7.4	6.2	3.9	2.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	7.3	6.2	3.9	2.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	7.3	6.3	3.9	2.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	7.3	6.3	3.9	2.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	7.4	6.2	3.9	2.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.3	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	1.2	1.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.3	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.2	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.2	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.2	1.1	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.2	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.2	1.1	0.4	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	7.5	22.5	39.4	19.4	1.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.2	8.6	20.4	38.3	15.7	1.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	7.7	21.7	39.8	19.3	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.2	8.8	20.4	37.4	16.2	2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	8.6	21.1	40.2	19.2	1.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.3	9.2	18.7	39.3	15.7	1.8
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	7.9	21.9	40.1	19.5	1.8
U3 - Efficacy additive alpha prior Beta(1, 1)	14.8	9.3	20	37.7	16.4	1.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.1	6.5	8	6.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	4.2	6.5	7.9	6.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.2	6.4	8	6.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	4.2	6.4	7.9	6.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.2	6.5	7.9	6.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.2	6.4	8	6.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.2	6.4	7.9	6.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.2	6.4	7.9	6.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.3	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	29.5	29.4	20	11.8	4.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	35.3	26.2	17.7	7	4.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	30	28.8	20.7	11.3	4.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.6	36	26.4	17.1	6.7	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	29.9	29	20.6	11.2	4.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	35.7	25.1	18.6	6.8	4.2
U2 - Efficacy additive alpha prior Beta(1, 1)	4.3	29	28.7	21.5	12.3	4.2
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	34.9	25.5	18	7	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	6.1	6.6	5.8	6.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.9	6.7	5.8	6.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	6.1	6.7	5.7	6.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	6	6.7	5.8	6.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.8	6.1	6.7	5.8	6.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	6	6.6	5.8	6.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.9	5.9	6.7	5.8	6.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.8	6.1	6.6	5.8	6.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.2	1.7	2.2	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.7	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.2	1.7	2.2	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	69	12.4	3.5	1	0.4	13.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	69.1	12.6	2.8	1.3	0.5	13.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.6	12.2	3.9	1.3	0.6	13.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.7	11.9	2.3	0.9	0.4	13.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.9	12.9	3.3	0.9	0.6	13.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	70	11.9	3	1.1	0.4	13.6
U2 - Efficacy additive alpha prior Beta(1, 1)	68.3	13.1	3.4	1.1	0.6	13.5
U3 - Efficacy additive alpha prior Beta(1, 1)	69.8	12.6	2.8	0.9	0.4	13.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.1	4.3	2.4	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.9	4.4	2.5	1.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.1	4.2	2.5	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19	4.3	2.4	1.2	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.1	4.4	2.4	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	4.2	2.5	1.2	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.1	4.2	2.4	1.2	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.1	4.3	2.5	1.2	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.5	0.3	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.5	0.4	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.7	34.2	21.4	10.9	4.4	2.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.6	37.6	21.3	8.9	2.3	2.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.9	34.2	22.1	10.8	3.8	2.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.1	37	20.4	9.8	2.5	2.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	34	22.9	10.5	3.8	2.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.8	37	20.4	8.6	3	2.2
U2 - Efficacy additive alpha prior Beta(1, 1)	27	32.9	23.1	10.8	3.9	2.3
U3 - Efficacy additive alpha prior Beta(1, 1)	28.3	35.9	21.9	9.2	2.5	2.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.5	6.7	5.8	3.9	2.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.4	6.8	6	3.9	2.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	6.8	5.8	3.9	2.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	6.8	5.8	3.9	2.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.4	6.8	5.8	3.9	2.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	6.8	5.8	3.9	2.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	6.8	5.9	3.9	2.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.4	6.8	5.9	3.9	2.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.1	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.5	1.1	1	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	1.1	1	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.5	1.1	1	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.3	8.1	2.5	1.5	0.4	43.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.7	7.2	1.7	1.1	0.4	42.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	43.8	7.9	2.8	1.3	0.3	43.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.8	7.9	1.7	1.2	0.2	43.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	43.8	9	2.2	1.3	0.6	43.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	44.9	8.9	1.7	0.9	0.3	43.3
U2 - Efficacy additive alpha prior Beta(1, 1)	43.7	8.7	2.7	1.2	0.5	43.2
U3 - Efficacy additive alpha prior Beta(1, 1)	45.1	8.2	2.1	1	0.3	43.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.8	4	2.1	0.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.8	4	2.2	0.8	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.9	3.9	2.1	0.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.7	4	2.2	0.8	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.8	4	2.2	0.8	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.9	3.9	2.2	0.8	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.8	4	2.2	0.8	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.9	4	2.1	0.8	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.7	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.6	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	1.7	0.4	0.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.5	0.2	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.6	0.4	0.2	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	10.6	15.9	11	27	18.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.7	11.9	14.4	10.3	23.3	18.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16	10.8	16.2	11.5	27	18.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.6	11.7	15.3	9.8	23.6	18
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	11	16.2	11.6	26.4	18.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.2	11.8	13.8	10.5	22.5	18.2
U2 - Efficacy additive alpha prior Beta(1, 1)	16.1	10	16.5	12	26.7	18.7
U3 - Efficacy additive alpha prior Beta(1, 1)	21.8	11.7	14	11.4	22.6	18.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	4.4	5.4	3.8	5.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.7	4.4	5.3	3.9	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	4.4	5.2	3.9	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.7	4.3	5.4	3.8	5.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	4.4	5.3	3.9	5.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	4.4	5.4	3.8	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.6	4.4	5.5	3.9	5.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.6	4.4	5.4	3.9	5.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.6	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.6	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.6	1.3	1.4	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.6	1.3	1.4	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.6	1.3	1.4	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	28.9	12.2	6.1	2.7	36.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.8	30.2	11.2	5.7	2.2	36.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	29.6	12.4	6.1	2.4	36.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	30	12	5.6	1.9	37.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	29.6	11.8	6.8	2.7	37
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14	30.4	11.4	5.6	2	36.6
U2 - Efficacy additive alpha prior Beta(1, 1)	12.3	29.9	12.2	6.3	2.4	36.9
U3 - Efficacy additive alpha prior Beta(1, 1)	13.3	30.5	12.2	5.4	1.9	36.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	6.3	4.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	6.2	4.2	2.4	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	6.2	4.2	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	6.2	4.1	2.3	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	6.2	4.2	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	6.2	4.2	2.4	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	6.2	4.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.2	6.3	4.1	2.4	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.3	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.3	0.8	0.6	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.8	0.6	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.3	0.8	0.6	0.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.1	20.4	3.7	0.6	0	7.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	72.4	17.7	2.5	0.4	0	7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.1	20.1	4.1	0.6	0	7.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	71.5	18.6	2.4	0.5	0	7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.1	20.2	4.2	0.6	0	6.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	71.9	18	2.8	0.3	0	7
U2 - Efficacy additive alpha prior Beta(1, 1)	66.9	21	4.5	0.6	0	7
U3 - Efficacy additive alpha prior Beta(1, 1)	70.8	19.4	2.4	0.5	0	6.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	6.2	3.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	6.2	3.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.9	6.1	3.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.8	6.1	3.5	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.9	6.1	3.5	1.2	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.9	6.2	3.4	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.9	6.2	3.5	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.7	6.3	3.6	1.1	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.4	0.7	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.7	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.4	0.7	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.4	0.8	0.4	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	13.2	10.5	32.2	22.7	3.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	24.2	13.4	8.1	31.2	19.9	3.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18	12.9	9.9	32.7	23.3	3.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.4	14	7.7	32.1	19.5	3.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.1	13.7	10.4	32	22.6	3.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.8	14.1	8	30.4	19.5	3.2
U2 - Efficacy additive alpha prior Beta(1, 1)	17.9	13.3	10.1	32.7	22.7	3.3
U3 - Efficacy additive alpha prior Beta(1, 1)	25	13.3	7.7	31.4	19.4	3.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5	4.6	5.2	7.2	7.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	4.6	5.2	7.2	7.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	4.6	5.2	7.2	7.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	4.6	5.2	7.3	7.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	4.6	5.2	7.1	7.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	4.7	5.2	7	7.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	5	4.6	5.3	7.2	7.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	4.6	5.3	7.1	7.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.4	2.2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.4	2.2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.9	1.4	2.2	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.5	29.8	13.5	5.9	2.5	31.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.8	31.7	11.6	5.2	1.7	32
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	29.8	14.4	5.8	2	31.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	32.2	12.5	4.7	1.7	32.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	30.1	13.5	5.9	2.1	32.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	31.1	12.9	5	1.5	31.9
U2 - Efficacy additive alpha prior Beta(1, 1)	16.8	29.7	13.3	6.1	2.1	32
U3 - Efficacy additive alpha prior Beta(1, 1)	17.3	31.7	12.4	4.8	1.9	31.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	6.3	4.5	2.7	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	6.3	4.5	2.7	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	6.3	4.6	2.7	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.9	6.4	4.6	2.6	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	6.3	4.5	2.8	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.9	6.3	4.6	2.7	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.9	6.4	4.5	2.7	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	6.4	4.5	2.7	1.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	0.9	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	0.9	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	0.9	0.8	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	0.9	0.8	0.3	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	15.5	13.5	24.6	16.4	7.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.6	16.6	10.2	21.3	12	7.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.3	15.3	14	24.6	16.3	7.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.4	16.9	9.5	22.5	11.6	7.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.8	14.8	13.7	26.6	15.9	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.6	15.4	10.7	22	12.9	7.4
U2 - Efficacy additive alpha prior Beta(1, 1)	22.5	14.8	14.7	24.5	16.3	7.2
U3 - Efficacy additive alpha prior Beta(1, 1)	32.3	15.2	11.5	21.9	11.8	7.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	5.3	5.6	5.9	5.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.9	5.2	5.5	6	5.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	5.2	5.5	5.9	5.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	5.2	5.6	6	5.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.9	5.2	5.5	6.1	5.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.9	5.1	5.5	5.9	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.8	5.2	5.6	5.9	5.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.9	5.2	5.5	5.9	5.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.1	1.4	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3	1.4	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1.4	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.4	2	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.4	2	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.1	1.4	2	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.5	2	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.1	1.4	2	0.8	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	52.1	26.1	10.1	3	1.7	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	62.6	22.4	5.9	1.3	0.8	7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	52.3	25.8	10.5	3.1	1.3	7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	63.4	22.2	5.4	1.5	0.5	7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	51.8	26.5	10.3	2.9	1.5	7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	63	22.7	5.5	1	0.8	7
U2 - Efficacy additive alpha prior Beta(1, 1)	53.1	25.1	10.2	3.1	1.5	7
U3 - Efficacy additive alpha prior Beta(1, 1)	63.6	22	5.2	1.5	0.6	7.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.1	7	5.7	3.4	1.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	7	5.6	3.4	1.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.1	7	5.6	3.5	2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.2	7.1	5.5	3.5	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	6.9	5.8	3.4	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.2	6.9	5.7	3.5	1.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	6.9	5.7	3.4	1.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.3	7	5.6	3.5	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	3.4	1.5	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.4	1.5	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.4	1.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.4	1.5	1.4	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	3.4	1.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.4	1.5	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.4	1.5	1.4	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.2	3.4	1.5	1.4	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.3	24.3	18.2	15.3	11.8	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	39.4	28.2	15.2	11.2	5.9	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.4	24.6	18.9	14.9	12.1	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	37.2	28.4	17.1	11.4	5.8	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.6	22.8	18.9	15.7	11.9	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	37.2	27.7	16.6	12	6.4	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	29.4	24.2	18.4	15.8	12.1	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	37.3	28.9	15.7	11.8	6.2	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6	5.5	5.4	5.8	7.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.1	5.4	5.4	5.7	7.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.1	5.5	5.4	5.6	7.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6	5.5	5.3	5.7	7.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	5.5	5.4	5.6	7.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6	5.5	5.4	5.7	7.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.9	5.4	5.4	5.8	7.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	6	5.5	5.4	5.8	7.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.2	1.8	2.6	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.2	1.8	2.6	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.2	1.8	2.6	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.1	1.8	2.7	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.2	1.8	2.6	1.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.7	48.1	17.2	1.7	0	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	34.7	49.1	14.6	1.3	0	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	31.3	48.3	18.5	1.6	0	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.3	48.5	14.7	1.2	0	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	32.2	47.5	18.6	1.4	0	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	34.9	49.1	14.6	1.1	0	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	33.2	48.1	17	1.4	0	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	35.1	48.5	15.3	0.8	0	0.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	9.7	7.2	2.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.9	9.8	7.2	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	9.9	7.2	2.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.9	9.8	7.3	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	9.6	7.2	2.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.9	9.8	7.1	2.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.9	9.7	7.2	2.6	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.9	9.8	7.2	2.6	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	1.2	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.8	1.1	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	8.4	11.2	14.3	55.7	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.9	11.4	10	12.9	48.1	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	7.9	10.7	14.9	55.3	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	12.4	10.6	12.5	47.7	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	8.7	10.4	15.3	55.4	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.8	11.9	11	12.3	47.3	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	8.7	11.2	15.6	54.9	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	15.8	11.9	9.9	12.8	48.9	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.8	4.7	5.4	12.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.8	3.7	4.6	5.5	12.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.7	4.6	5.4	12.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.8	3.8	4.6	5.4	12.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.7	4.7	5.4	12.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	3.8	4.6	5.5	12.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.8	4.6	5.4	12.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	3.7	4.6	5.4	12.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	50.3	16.6	2	0	8.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.1	50.7	13	1.7	0	8.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.6	49.3	16.1	2.6	0	8.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.9	49.7	13.7	1.4	0	8.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.8	50.9	16.9	2.1	0	8.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.1	50.4	13.8	1.4	0	8.3
U2 - Efficacy additive alpha prior Beta(1, 1)	23	51	15.9	2	0	8.1
U3 - Efficacy additive alpha prior Beta(1, 1)	25.9	51.3	13	1.5	0	8.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.9	10	7.2	2.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	10.2	7.1	2.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.2	7.2	2.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	10.3	7.1	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9	9.9	7.2	2.5	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	10	7.2	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.8	10.2	7.1	2.4	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.9	10.1	7.1	2.5	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.3	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.5	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.5	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.5	1.3	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	1.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.5	1.3	0.8	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	14	43.4	28.2	8.7	1.4	4.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.9	46.5	22.5	7.1	0.7	4.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	44.3	28.1	8.8	1.3	4.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19	47.1	22	7	0.6	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.5	43.2	27.9	9.7	1.4	4.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.5	45.8	22.2	7.4	0.8	4.3
U2 - Efficacy additive alpha prior Beta(1, 1)	13.4	45.2	27.8	7.9	1.4	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	45.3	22.7	7.2	0.8	4.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	8.3	8.3	4.6	2.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.7	8.3	8	4.9	2.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	8.3	8.2	4.8	2.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	8.4	8.2	4.8	2.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	8.4	8.2	4.7	2.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	8.3	8.2	4.8	2.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.7	8.2	8.2	4.8	2.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.7	8.4	8.2	4.7	2.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.5	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.6	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.5	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.5	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.5	1.8	1.6	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.5	1.8	1.6	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.5	1.8	1.6	0.5	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	59	1	0	0	0	40
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	59.1	1.2	0	0	0	39.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	59.5	1	0.1	0	0	39.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	59.8	1	0	0	0	39.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.9	1.2	0	0	0	39.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	59.4	1.1	0	0	0	39.5
U2 - Efficacy additive alpha prior Beta(1, 1)	59.4	1.2	0	0	0	39.4
U3 - Efficacy additive alpha prior Beta(1, 1)	59.8	0.8	0	0	0	39.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.5	2.3	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.6	2.2	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.4	2.4	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.5	2.3	0.6	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	2.2	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.5	2.3	0.7	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20.5	2.3	0.6	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.5	2.3	0.7	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	0.7	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	0.7	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	28	52.2	13.5	1.7	0	4.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	32.7	50.9	10.4	1.4	0	4.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.8	53	13.1	1.5	0.1	4.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.3	51.1	9.4	1.5	0	4.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.2	53.3	12.1	1.7	0.1	4.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	31.9	52	10	1.4	0	4.7
U2 - Efficacy additive alpha prior Beta(1, 1)	29.5	50.9	13.2	1.7	0	4.7
U3 - Efficacy additive alpha prior Beta(1, 1)	32.7	51.9	9.6	1.3	0	4.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	11.9	6.5	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.8	11.8	6.4	2	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	12	6.5	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	12	6.3	1.9	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	11.9	6.4	1.9	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.8	11.9	6.5	1.9	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.7	11.8	6.5	2	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.8	11.9	6.5	1.9	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.9	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.8	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.8	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.9	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.8	1.2	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.9	1.3	0.5	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.2	0.5	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.6	4.1	0.4	0	0	73.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.2	3.8	0.1	0	0	74.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.2	4.2	0.1	0	0	74.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.2	3.7	0	0	0	74.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.6	3.7	0.3	0	0	74.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22	3.4	0	0	0	74.6
U2 - Efficacy additive alpha prior Beta(1, 1)	21.4	3.7	0.2	0	0	74.7
U3 - Efficacy additive alpha prior Beta(1, 1)	21.2	4.1	0	0	0	74.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	3.2	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.6	3.1	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	3.2	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.6	3.1	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.6	3.3	1	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.7	3.1	1	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	16.6	3.3	0.9	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	3.1	1	0.1	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	1.4	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	1.5	0.2	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	1.5	0.2	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	1.5	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.6	1.1	6.7	42.7	47.8	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.7	2.4	7.6	45	42.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.4	1.3	6	44.8	46.4	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2.8	2.6	6.7	45.5	42.3	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	1.3	5.8	45	46.5	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	2.9	2.4	7.3	45.4	42	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.3	1	6.4	44	47.3	0
U3 - Efficacy additive alpha prior Beta(1, 1)	2.8	2.4	7.1	44.8	42.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.2	3.7	8.2	11.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.1	3.7	8.2	11.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.7	8.2	11.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	3.2	3.7	8.2	11.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.1	3.7	8.3	11.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.3	3.1	3.7	8.2	11.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.3	3.1	3.7	8.1	11.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.2	3.1	3.8	8.2	11.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.2	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.5	28.7	29.6	4.5	0	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	40.3	30.5	24.8	3.7	0	0.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.3	29.4	29	4.6	0	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	40.7	28.8	26	3.8	0	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	35	28.6	31.3	4.4	0	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	41.4	29.1	25.6	3.2	0	0.7
U2 - Efficacy additive alpha prior Beta(1, 1)	35.8	29.9	29.1	4.5	0	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	42.3	28.6	24.6	3.8	0	0.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	8.2	8.8	3	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	8	9.1	2.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	8	8.9	3	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	8	9	2.9	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	8.1	9	2.9	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	8	8.9	3	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.3	8	9	3	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.4	7.9	8.9	3.1	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	3.1	1.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3	1.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3	1.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.1	1.5	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3	1.5	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	1.5	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3	1.5	0.9	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.3	26.9	24.8	4.8	0.1	10.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	36.1	26.9	21.9	4.8	0.1	10.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	32	27	26	4.7	0.1	10.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.1	26.8	22.6	4.3	0	10.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.1	26.9	24.9	4.8	0.1	10.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	36.6	26	23	4.2	0	10.2
U2 - Efficacy additive alpha prior Beta(1, 1)	32.1	27.8	25.1	4.8	0.1	10.1
U3 - Efficacy additive alpha prior Beta(1, 1)	36	27	22	4.9	0	10.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	6.8	7.8	3.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.8	6.7	7.9	3	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	6.7	7.9	3	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	6.8	7.8	3	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	6.8	7.7	3.1	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	6.8	7.8	3	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.8	6.7	7.9	3	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.7	6.7	8	3	0.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3	1.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.3	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3	1.2	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.9	1.2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3	1.3	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3	1.2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3	1.3	0.9	0.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23	24	40	6.7	0.9	5.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.9	24.7	36.4	6.2	0.4	5.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.5	24.5	39.8	7.1	0.7	5.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27	25.5	35.1	6.4	0.6	5.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.6	22.8	40.8	6.6	0.8	5.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.5	24.9	35.3	5.3	0.6	5.4
U2 - Efficacy additive alpha prior Beta(1, 1)	22.8	24.7	38.8	7.7	0.6	5.4
U3 - Efficacy additive alpha prior Beta(1, 1)	28.1	23	36.9	6.1	0.4	5.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.1	6.5	9.2	4.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.1	6.4	9.3	4.1	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.1	6.5	9.1	4.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	6.6	9.1	4.2	1.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	6.6	9.1	4.2	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	6.5	9.2	4.2	1.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.1	6.6	9	4.3	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.1	6.5	9.1	4.1	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.9	1.7	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.9	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.9	1.6	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.6	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	2.9	1.6	1.1	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.7	34.6	6.9	0.1	0	32.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.6	35.7	5.5	0.1	0	33.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.4	34	6.9	0.1	0	32.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.2	35.2	5.5	0.1	0	33
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	33.8	7.5	0	0	32.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	24.1	37	6.1	0.2	0	32.6
U2 - Efficacy additive alpha prior Beta(1, 1)	25.7	34.5	7	0.1	0	32.7
U3 - Efficacy additive alpha prior Beta(1, 1)	25.8	35.1	5.7	0.4	0	33
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	8.9	4.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	8.9	4.2	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	12	8.7	4.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	8.8	4.2	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	8.9	4.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	8.8	4.2	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	12	8.7	4.2	1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12	8.7	4.2	1	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	0.8	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	0.8	0.3	0	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.8	22.5	3.8	0.1	0	2.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	72.8	21.6	2.7	0.2	0	2.7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	70	23.1	3.8	0.2	0	2.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	72.9	21.8	2.5	0.1	0	2.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.3	23.1	3.5	0.2	0	2.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.6	21.1	2.4	0.2	0	2.7
U2 - Efficacy additive alpha prior Beta(1, 1)	70.2	23.3	3.6	0.1	0	2.8
U3 - Efficacy additive alpha prior Beta(1, 1)	73.1	20.6	3.2	0.2	0	2.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.9	7.5	3.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	7.6	3.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	7.7	3.5	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	7.7	3.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.6	7.6	3.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.7	7.6	3.5	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.7	7.7	3.5	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.7	7.6	3.5	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.1	0.8	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.1	0.7	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.1	0.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	2	0.7	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.1	15.8	14.1	35.3	11.9	0.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.2	18.8	12.1	29.9	10.2	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.1	14.5	15.3	35.3	13	0.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.9	17	12.7	30.4	10.2	0.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.2	13.6	14.6	35.3	12.5	0.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.3	18.5	13.5	29.3	10.6	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	21.6	15.3	15.1	34.6	12.6	0.8
U3 - Efficacy additive alpha prior Beta(1, 1)	28.4	18.6	12.8	30	9.4	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.1	4.6	5.8	8.1	6.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.5	5.8	8.1	6.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	4.5	5.8	8.1	6.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.5	5.7	8	6.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.2	4.5	5.8	8.1	6.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	5	4.5	5.9	8.1	6.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	4.5	5.8	8.2	6.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	2.8	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	1.6	2.3	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.7	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.8	1.6	2.2	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	2.7	1.6	2.3	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	1.6	2.3	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	2.7	1.6	2.2	0.9	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	31	28.4	25.8	7.3	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	37.4	26.8	21.5	4.5	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.9	31.4	28.1	24.9	6.6	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	37.9	27.1	21.1	4.4	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.1	30.7	30.1	24.3	6.7	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	38.9	26.4	20.9	4.3	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	8.1	30.9	29.3	25.2	6.4	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	9.8	38.1	27.6	19.9	4.5	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.5	6.3	8	6.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	5.5	6.4	7.9	6.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	5.5	6.4	7.9	6.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	5.4	6.5	7.9	6.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.4	6.5	7.9	6.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	5.5	6.4	7.9	6.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.4	6.4	7.9	6.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.1	5.5	6.5	7.8	6.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.7	1.4	1.8	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.8	1.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.4	1.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.7	1.4	1.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.7	1.3	1.8	1.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.7	1.3	1.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.4	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	14.3	34.9	25.9	14.2	0.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.9	18.6	31.4	23.3	8.2	0.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.3	14	35.2	25.6	14.3	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.8	20.5	31.1	23.2	7.8	0.6
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	14.3	34.5	25.1	14.5	0.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.5	18.2	32	24.2	8.5	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	13.4	35	26	13.8	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	16.6	19	31.9	23.3	8.6	0.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	4.3	7.5	7	7.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	4.2	7.4	7.1	7.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.3	7.4	7.1	7.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4	4.2	7.4	7.1	7.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	4.3	7.3	7.1	7.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.1	4.2	7.4	7	7.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	4.2	7.5	7	7.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4	4.2	7.4	7.1	7.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.8	1.9	2.6	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	2.8	1.9	2.6	1.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.6	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.6	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.9	2.6	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.8	1.9	2.6	1.1	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	5.8	13.8	38.8	33.1	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	14.5	7.5	11.7	39.1	27.2	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	4.7	13.4	39.3	32.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	7.6	12.9	39	27.1	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	5.5	13.4	39.6	32.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.4	6.8	12.6	38	28.2	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9	6.4	13.1	39.3	32.2	0
U3 - Efficacy additive alpha prior Beta(1, 1)	13.9	7.6	12.9	38.1	27.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.6	5	8	9.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	3.5	5	8.2	9.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	3.5	5.1	8.1	9.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.5	5.1	8.1	9.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.5	5	8.2	9.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.5	5.1	8.1	9.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.5	5	8.1	9.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.5	5.1	8.1	9.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.7	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.7	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.8	1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.6	1.4	2.7	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.7	13.8	44.5	17.2	6	1.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.8	15.8	39.2	15.6	3.6	2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.5	14.5	43.9	16.9	6.3	1.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.2	15.6	39.6	14.9	3.9	1.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17	13.9	44.8	16.8	5.7	1.8
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.5	16.9	40.2	14.5	3.9	2
U2 - Efficacy additive alpha prior Beta(1, 1)	15.9	14.2	45.6	15.7	6.6	2
U3 - Efficacy additive alpha prior Beta(1, 1)	23.9	15.9	39.5	15.2	3.6	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.7	9.3	5.6	3.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.3	5.7	9.5	5.6	3.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	5.6	9.5	5.6	3.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.2	5.7	9.5	5.6	3.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.7	9.5	5.6	3.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	5.8	9.4	5.5	3.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.3	5.7	9.5	5.5	3.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.3	5.6	9.6	5.5	3.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.1	1.9	1.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.9	1.6	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.2	1.9	1.7	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.9	1.7	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.2	1.9	1.7	0.6	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	36.4	36.6	11.4	3.6	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	42	30.2	9.7	2.2	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	37.9	34.9	12.2	3.5	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.1	43.3	29.4	9	2.1	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.5	37	35.7	12.1	3.6	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.3	42.5	28.6	10	2.5	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	12.4	38.7	34.4	10.8	3.6	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	42.6	29.7	9.7	2.4	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	8	9	5.3	3.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	7.9	9	5.2	3.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	8	8.9	5.3	3.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	8	9	5.3	3.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	7.9	9	5.3	3.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.6	8	8.9	5.3	3.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.6	8	8.9	5.3	3.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.6	8	8.9	5.3	3.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.6	1.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3	1.6	1.4	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.6	1.4	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3	1.6	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.4	38.1	14.6	6.6	2.6	14.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.7	36.7	11.2	6	1.4	15
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.5	38.2	14.2	6.3	2.8	15
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.6	38.8	10.7	5.6	1.6	14.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.1	38.1	14.7	6.3	3.1	14.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.5	37	12.1	5.8	2	14.6
U2 - Efficacy additive alpha prior Beta(1, 1)	23.2	37.1	15.5	6.5	2.8	14.9
U3 - Efficacy additive alpha prior Beta(1, 1)	28	37.3	12.5	6	1.5	14.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	8.4	5.7	3.1	1.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	8.3	5.7	3.2	1.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	8.4	5.7	3.1	1.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	8.4	5.6	3.2	1.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	8.4	5.6	3.1	1.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	8.3	5.5	3.1	2	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.2	8.5	5.6	3.1	1.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	8.2	5.8	3.2	1.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.5	1.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.4	1.1	1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.5	1.1	1.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.4	1.2	1	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.4	1.1	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.5	1.1	1	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.4	1.2	1.1	0.3	

Phase I/II Operating Characteristics

Random model - Efficacy additive alpha prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.5	47.1	12.9	2.6	0.9	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	36	44.4	9.8	2.1	0.7	7
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.9	48.3	13.6	2.5	0.9	6.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	36.9	44.4	9.5	1.8	0.6	6.8
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.1	46.6	12.9	3.2	1	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	37.3	43.4	9.7	2.3	0.4	6.9
U2 - Efficacy additive alpha prior Beta(1, 1)	29.1	46	14.4	2.8	0.9	6.8
U3 - Efficacy additive alpha prior Beta(1, 1)	35.5	45.6	9.3	2	0.8	6.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.4	10.1	6.5	2.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.5	10.2	6.4	2.6	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	10.1	6.5	2.6	1.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.4	10.2	6.4	2.7	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	10.3	6.3	2.7	1.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.4	10.2	6.4	2.7	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.5	10.1	6.5	2.6	1.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.4	10.2	6.4	2.6	1.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.3	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4	1.3	1	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4	1.3	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.3	1	0.2	